ContextWorld: Benchmarking In-Context Dynamics Adaptation in Latent Visual World Models

Abstract

Latent world models predict future states from visual observations and action sequences within a learned representation space. When environment dynamics vary, accurate prediction requires inferring hidden dynamics from interaction history. However, because models can exploit immediate visual shortcuts or regress to average dynamics to achieve low loss without history conditioning, conventional prediction accuracy fails to reveal genuine adaptation. We introduce ContextWorld, a benchmark designed to evaluate in-context dynamics adaptation in action-conditioned latent world models under strictly frozen weights. ContextWorld employs paired counterfactual evaluations—varying interaction histories while holding the current observation and future action sequence fixed—to concurrently quantify adaptation accuracy (whether predictions correctly distinguish between diverging future states) and history sensitivity (whether matching history yields lower prediction error than mismatched history). Spanning nine tasks across four environments, the benchmark provides separate training and evaluation splits with continuous interaction rollouts under differing dynamics. Our systematic evaluation of representative latent world models (LeWM, PLDM, and DINO-WM) yields three key findings: First, models trained under fixed dynamics make negligible use of interaction history across most tasks. Second, mixing multi-dynamics data improves history responsiveness, but the benefits of data scaling and the relative effectiveness of training strategies vary substantially across tasks. Third, under certain training schemes, models fail to effectively leverage history even on the training distribution, despite steadily decreasing training loss. These results reveal a notable mismatch between prediction-loss reduction and in-context dynamics adaptation, suggesting that standard prediction objectives do not reliably drive models to use interaction history and highlighting the need for objectives and benchmarks explicitly targeted at context-driven physical reasoning.

1. Introduction

Visual world models learn environment dynamics to predict future states [19], with latent world models performing these predictions within learned representations of visual observations [20, 21]. A model trained under fixed dynamics can achieve high prediction accuracy without utilizing interaction history to adapt its predictions to current conditions. However, such predictions can fail when environment dynamics vary [33]. For example, visually identical objects can move different distances under the same push if surface friction differs. Interaction history provides evidence about these hidden physical properties. We investigate whether latent world models can leverage this contextual evidence to predict future states under varied dynamics *without updating their weights*—an ability we term *in-context dynamics adaptation*.

Physical prediction benchmarks evaluate whether observed interactions help models infer hidden properties or predict downstream outcomes governed by them [4, 32]. However, benchmarks such as Physion++ evaluate visual models through auxiliary readout classifiers [32]. We introduce ContextWorld to evaluate in-context dynamics adaptation directly within action-conditioned latent world models, leveraging their existing latent predictors without training separate classifiers for future events. The benchmark spans nine tasks across four environments, covering physical parameter variations and interaction rules, and provides separate training and evaluation data.

In its paired tasks, two dynamics conditions share the identical current observation and future action sequence, but produce diverging ground-truth futures. Interaction history is therefore strictly necessary to distinguish the two outcomes. Because a deterministic model that ignores history receives identical inputs and yields identical predictions for both conditions, it scores at most 50% on paired future selection when ties count as failures. We evaluate adaptation accuracy by testing whether predictions distinguish the true futures under different dynamics. We assess history sensitivity by comparing prediction errors with matching and mismatched histories.

Using ContextWorld, we benchmark three representative latent world models: LeWM [24], PLDM [30], and DINO-

WM [37]. We find that after training under fixed dynamics, these models make negligible use of interaction history across most tasks. To examine how this adaptation ability can be acquired, we mix ContextWorld training data with original rollouts and compare training from scratch against fine-tuning models pretrained under fixed dynamics. We train models separately for each task; DINO-WM retains its frozen pretrained visual encoder in both settings. We further scale up the training data and examine the impact of freezing the visual encoder in LeWM and PLDM. Our experiments reveal that data scaling improves history utilization on several tasks, while the most effective model and training strategy vary substantially by task. For example, starting from the same pretrained weights, freezing PLDM's visual encoder improves its ability to distinguish future states under varying contact friction, but reduces this ability under motion damping.

To understand why models often fail to leverage history, we analyze model predictions on the training distribution itself. Under certain training schemes, training loss steadily decreases even while models make limited use of interaction history to distinguish dynamics conditions. Because this failure occurs on the training data, it cannot be explained by out-of-distribution generalization alone, suggesting that standard prediction objectives may not sufficiently encourage models to use interaction history for prediction.

Our contributions are:

1. **The ContextWorld benchmark.** We introduce nine hidden-dynamics tasks across four environments, with separate training and evaluation data. Paired counterfactual evaluations measure future discrimination and the benefit of matching history directly from latent predictions, without weight updates or auxiliary readouts.
2. **Systematic empirical comparison.** We benchmark representative latent world models across data scales, training strategies (scratch vs. fine-tuning), and visual encoder freezing schemes, demonstrating that scaling benefits and optimal adaptation strategies depend heavily on the underlying physical regime.
3. **Empirical evidence of adaptation failures during training.** We show that training loss can steadily decrease even when models fail to leverage interaction history on the training data, providing empirical evidence that standard prediction losses do not reliably drive in-context dynamics adaptation.

2. Related Work

Visual world models. Visual world models learn environment dynamics to simulate and predict future states. Early frameworks [19] combined learned visual representations with recurrent dynamics to support internal policy training. Subsequent methods advanced this paradigm through online planning (PlaNet [20]) and policy learning via imagined latent trajectories (Dreamer [21]). While these foundational approaches rely on pixel-level image reconstruction to learn visual representations, recent Joint-Embedding Predictive Architectures (JEPAs) [23] shift the training objective toward predicting target representations directly without reconstructing pixels. This reconstruction-free principle has been applied to masked spatial regions (I-JEPA [2]) and extended to temporal video representation learning (V-JEPA [5]). To support physical interaction and decision-making, models like V-JEPA 2 [3] introduce an action-conditioned predictor trained on robot interactions atop pretrained video representations.

Action-conditioned latent world models also differ in how they learn visual representations. DINO-WM [37] trains a predictive dynamics model on top of strictly frozen pretrained features for planning from offline trajectories. In contrast, PLDM [30] jointly optimizes the visual encoder and the predictor using reward-free offline data. Addressing the inherent instabilities of joint training, LeWM [24] streamlines the objective into a pure latent prediction loss paired with a single representation regularizer.

In-context adaptation. When environment dynamics shift between training and deployment, interaction history provides a critical signal for adaptation without requiring costly weight updates. Early approaches in meta-reinforcement learning [6] leveraged this principle by training agents to adapt from limited experience across tasks. Specifically, RL^2 [14] implicitly stores interaction history within a recurrent network's hidden state, while PEARL [29] explicitly infers a task representation from experience to condition the policy. Paralleling the success of large language models, which learn from in-context examples without parameter updates [8, 13], this adaptation paradigm has increasingly been extended to visual world models.

Within world models, contextual history is utilized to guide predictions of action outcomes under novel conditions. For instance, OSWM [16] pretrains a transformer on synthetic dynamics, using observed state transitions as context to forecast futures in new environments. L2World [33] investigates the mechanics of this adaptation, distinguishing between using context to identify previously seen environments versus learning entirely new dynamics on the fly. It systematically studies the effects of context length and training environment diversity through multi-step prediction errors and context perturbations. For robotic control, ICWM [34] employs task-agnostic interaction history to adapt policies to novel camera viewpoints without weight updates, using ablations over correct, absent, and mismatched interaction contexts to validate history reliance.

Benchmarks. To evaluate adaptation across diverse environments, existing benchmarks introduce controlled variations in dynamics, appearance, or underlying task rules. For instance, Meta-World [35] and Procgen [11] test generalization across manipulation tasks and procedurally generated levels. CausalWorld [1] and CARL [7] directly vary environment properties, while the XLand series (XLand-MiniGrid [26], XLand-100B [27]) evaluates policy adaptation from interaction history across distributions of tasks.

Within the domain of physical prediction, several benchmarks focus on counterfactual reasoning and hidden property inference. CoPhy [4] asks models to predict outcomes following changes to a scene's initial configuration, utilizing the original trajectory as evidence of hidden properties. Filtered-CoPhy [22] refines this by ensuring the observed trajectory contains sufficient information for the counterfactual prediction. ComPhy [9] probes reasoning regarding hidden mass and charge through predictive and counterfactual video querying, while Physion++ [32] tests whether interaction history aids in predicting future contact in paired trials that share the identical current observation but differ in physical properties. While these benchmarks provide critical insights, Physion++ assesses visual models via auxiliary readout classifiers rather than directly scoring their predicted representations [32].

Beyond visual domains, a recent time-series benchmark [36] reveals that low absolute prediction error can misleadingly coexist with incorrect response directions when actions change. A critical remaining question is whether and when modern action-conditioned latent world models learn to leverage interaction history for accurate, in-context physical prediction. ContextWorld addresses this gap by directly evaluating latent predictors through controlled comparisons across models, tasks, data scales, and training strategies.

3. ContextWorld Benchmark

ContextWorld is designed to evaluate in-context dynamics adaptation—specifically, whether models leverage interaction history to accurately predict future states conditioned on an action sequence. This section formalizes the prediction setting, describes how we construct the tasks and datasets, and explains the paired evaluation protocol and metrics.

3.1. Problem Formulation

To strictly isolate in-context adaptation from weight updates, all model parameters θ remain fixed during evaluation. The model receives exclusively visual observations and actions. For a target horizon τ, the model receives an interaction history H_t, comprising visual observations up to and including time t and the actions executed between them, alongside a planned future action sequence $A_{t:t+\tau-1}$. The prediction and its target are formulated as:

$$\begin{aligned} \hat{z}_{t+\tau} &= f_\theta(H_t, A_{t:t+\tau-1}), \\ z_{t+\tau} &= e_\theta(o_{t+\tau}). \end{aligned} \tag{1}$$

Here, $o_{t+\tau}$ is the ground-truth visual observation at time $t+\tau$, and e_θ is the model's native target encoder. The target encoder is also frozen during evaluation. We compare predictions directly with these encoded targets in latent space.

3.2. Task Design

To systematically evaluate dynamics adaptation, we construct nine tasks spanning four base environments by introducing controlled variations in physical parameters and underlying interaction rules. These environments build upon TwoRoom navigation [37], the PushT pushing environment [10], Reacher from the DeepMind Control Suite [31], and Cube from OGBench [28]. The tasks and their corresponding history lengths are summarized in Table 1.

To test whether models use interaction history for prediction, we construct paired samples for six tasks: pusher displacement, arm mass, contact friction, motion damping, gripper carry rule, and portal exit. The paired samples share the current observation and future action sequence, but have different hidden dynamics. For each sample, we generate the history and ground-truth future in one continuous simulation with fixed dynamics. The two futures differ, so accurate prediction requires information from the history. Construction checks are detailed in Appendix B.

The speed task tests whether models can use history to predict motion at unseen speeds. We hold the current observation and future actions fixed and provide histories recorded at different speeds. Action delay covers several delay values to test predictions before and after an action takes effect. For example, with delays of six or ten simulator steps, no movement occurs within the next five steps. These delays produce the same future state, so we score them as one response group and weight all groups equally (Appendix C). For the door task, we test whether the model correctly predicts passage or blockage from the interaction history.

3.3. Dataset

For each task, we provide Training, Development, and Test splits with no shared interaction trajectories. The Development split is used to select methods and checkpoints. These splits may share dynamics parameter values or interaction rules. The speed task also tests speeds not used in training, both within the training range (interpolation) and outside it (extrapolation).

The paired tasks provide 32,768 training pairs per task, except for gripper carry, which provides 10,000. Each has 256 Development pairs and 256 Test pairs. Appendix D

Table 1. The nine tasks test whether interaction history helps models predict the effects of hidden dynamics. History length h counts observation frames, including the current frame; sampling details are given in Appendix A.

Task	Environment	h	What history helps the model predict
Speed	TwoRoom	3	How far the agent moves under the same action
Pusher displacement	PushT	3	Whether the same action moves the pusher a shorter or longer distance
Arm mass	Reacher	3	How the same torque changes arm motion
Action delay	TwoRoom	7	When an action takes effect
Contact friction	PushT	3	How friction changes object motion during contact
Motion damping	PushT	3	How quickly object motion decays after contact ends
Gripper carry rule	Cube	3	Whether a closed gripper carries the cube
Door traversability	TwoRoom	3	Whether a visually identical door permits passage
Portal exit	TwoRoom	3	Where the agent emerges after entering the same portal

provides complete dataset statistics, dynamics settings, and split checks for all nine tasks.

3.4. Evaluation

A model can change its predictions with the interaction history and still predict the wrong future. We therefore evaluate whether predictions distinguish the true futures under different dynamics, measure prediction error, and compare errors with matching and mismatched histories. We report benchmark results on the Test split.

Prediction accuracy. All tasks except Speed and action delay use the same accuracy rule. For these tasks, we compare each prediction with both true futures. A prediction using the history from condition A is counted as correct only if it is closer to A's true future than to B's, and vice versa. For pair i, z_i^A and z_i^B denote the true future representations, and $\hat{z}_i^A$ and $\hat{z}_i^B$ denote the corresponding predictions. Using mean squared error d, we report accuracy as the percentage of correct predictions across both conditions in N pairs:

$$\begin{aligned} A_{\text{future}} &= \frac{100}{2N}\sum_{i=1}^{N}(I_i^A + I_i^B), \\ I_i^A &= \mathbf{1}\{d(\hat{z}_i^A, z_i^A) < d(\hat{z}_i^A, z_i^B)\}, \\ I_i^B &= \mathbf{1}\{d(\hat{z}_i^B, z_i^B) < d(\hat{z}_i^B, z_i^A)\}. \end{aligned} \tag{2}$$

Speed and action delay involve more than two dynamics conditions. For each tested speed, we compare predictions from different histories with the true future at that speed. We count the prediction from the matching history as correct only when its error is lower than that of every other prediction. For action delay, the prediction horizon is five simulator steps. Delays 0–4 form five separate response groups, while delays 5–10 form one group because the action has not yet taken effect. We assign each prediction to the group containing the closest true future, compute accuracy for each group, and report the mean across the six groups. Appendix E gives the detailed scoring rules.

Multi-step prediction error. High prediction accuracy does not guarantee low prediction error. We therefore evaluate predictions over multiple steps, using the same future action sequences for all dynamics conditions in each scene. Models roll out from the observed interaction history without receiving further ground-truth observations. We evaluate five successive predictions, up to 25 simulator steps ahead (Appendix E).

We compare prediction errors with a reference that makes the same prediction across dynamics conditions [25]. For each scene, action sequence, and prediction time, this reference is the mean of the true future representations. With equally weighted conditions, this mean gives the lowest squared error among predictions that do not distinguish the conditions [18]. For scene q, let z_{qkct} and $\hat{z}_{qkct}$ denote the true and predicted future representations for dynamics condition k, action sequence c, and prediction time t. The reference prediction is $\bar{z}_{qct} = K_q^{-1}\sum_k z_{qkct}$, where K_q is the number of dynamics conditions. We average the model's squared errors within each scene to obtain L_q, and the reference errors to obtain B_q. Both averages cover all K_q conditions, C_q action sequences, and $T = 5$ prediction times. We then report the ratio of the summed model errors to the summed reference errors:

$$\begin{aligned} L_q &= \frac{1}{K_q C_q T}\sum_{k,c,t}\|\hat{z}_{qkct} - z_{qkct}\|^2, \\ B_q &= \frac{1}{K_q C_q T}\sum_{k,c,t}\|z_{qkct} - \bar{z}_{qct}\|^2, \\ E/B &= \frac{\sum_q L_q}{\sum_q B_q}. \end{aligned} \tag{3}$$

Lower E/B is better: 0 means an exact prediction in the target representation, and 1 matches the error of the mean-future reference. Values below 1 outperform this reference; values above 1 indicate larger errors. To measure the benefit of matching history, we also compare errors with matching and mismatched histories.

History controls. To test whether matching history helps prediction, we perturb the input history [33]. For each sample, we replace the history with one from a different dynamics condition in the same scene. The current observation and future actions remain unchanged. We evaluate both predictions against the true future of the original sample. History (%) is the percentage of comparisons in which matching history gives strictly lower prediction error than mismatched history. Speed also uses this strict comparison as its task score.

History (%) measures how often matching history reduces error, but not by how much. We therefore also report the reduction relative to the same reference error used for E/B. Let E_+ and E_- be the summed scene errors with matching and mismatched histories, respectively, using the averaging over conditions, actions, and times in Eq. 3. For each sample, errors from multiple mismatched histories are averaged equally. With $B = \sum_q B_q > 0$, the normalized error reduction is

$$G_H = \frac{E_- - E_+}{B}. \tag{4}$$

A positive G_H means that matching history reduces error, but does not imply that the predictions are accurate. Their overall error is measured by E/B.

Conditional response error. Training loss can decrease even when the model does not accurately predict how different dynamics change the future. We therefore compare the difference between the two predicted futures with the difference between the two true futures. For pair i, let $\Delta z_i = z_i^B - z_i^A$ denote the true difference and $\Delta \hat{z}_i = \hat{z}_i^B - \hat{z}_i^A$ the predicted difference. We define normalized response error (NRE) as the summed squared error between these differences, divided by the summed squared magnitude of the true differences:

$$\mathrm{NRE} = \frac{\sum_i \|\Delta \hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}. \tag{5}$$

Lower NRE is better: 0 means that the predicted difference exactly matches the true difference. Predicting no difference gives NRE = 1, although other incorrect predictions can produce the same value. Errors shared by both predictions cancel when we subtract them, so a low NRE does not guarantee that either predicted future is accurate.

Comparing prediction errors. E/B and NRE measure errors in each checkpoint's target representation. Normalization does not make these representations equivalent, so lower values across models do not necessarily indicate more accurate physical predictions. Changes across training checkpoints can also reflect changes in the target encoder. A fixed target representation removes this source of variation.

Reporting results. We report the mean and standard deviation across training runs with the same settings, weighting each run equally. For multi-step errors, we also report confidence intervals that reflect variation across sampled scenes or source trajectories, rather than across training runs (Appendix E).

4. Experiments

4.1. Experimental Setup

To study how training data affect adaptation, we compare training on original-environment data alone with training on a mixture of original-environment and ContextWorld data. We use the original-environment datasets released with LeWM [24]. For each task, mixed training samples 50% of its trajectory windows from the corresponding original-environment dataset and 50% from the task's ContextWorld Training split.

We evaluate LeWM [24], PLDM [30], and DINO-WM [37] to examine how original-environment pretraining and visual encoder updates affect adaptation. Table 2 defines the four training schemes used in the comparisons. Original and Mixed each train for 10 epochs. The two Original $\rightarrow$ Mixed schemes start from the same Original checkpoint and train for another 10 epochs on mixed data, either updating or freezing the visual encoder. Each mixed-data model is trained for one task, so a row of nine task results represents separate task-specific checkpoints.

We also evaluate planning with the cross-entropy method (CEM) [12] in the original environments to test whether models retain their planning ability after training on mixed data. These scores measure performance in the original environments, not adaptation to hidden dynamics.

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Table 2. Training schemes. Original data come from the corresponding environment; mixed data use the 50/50 sampling described in the text. LeWM and PLDM start Original and Mixed from random weights. DINO-WM starts with a pretrained, frozen visual encoder and a randomly initialized predictor.

Scheme	Starting weights	Training data	Visual encoder	Other trainable modules
Original	Model initialization	Original only	Updated†	Updated
Mixed	Model initialization	Mixed	Updated†	Updated
Original → Mixed (Joint)	Original checkpoint	Mixed	Updated	Updated
Original → Mixed (Frozen Encoder)	Same Original checkpoint	Mixed	Frozen	Updated

†DINO-WM's visual encoder remains frozen in all experiments, so Original → Mixed (Joint) is not evaluated for this model. In Original → Mixed (Frozen Encoder), only the visual encoder is frozen; the remaining trainable modules, including any projector and action encoder, continue to update. All schemes retain the model's original training losses.

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A. Input Sampling

Observations are typically sampled every five simulator steps. The actions executed between consecutive observations are included in the interaction history.

B. Trajectory Construction and Pairing

Continuous trajectories. Each sample follows a continuous simulator trajectory from its history through its future, without intermediate resets or state overrides. Identical current observations may still conceal differences in velocity or contact state, so we perform task-specific checks of physical-state matching at the start of prediction. For action delay, we retain the actual queue of pending actions.

Starting positions. For tasks in which the two dynamics conditions cannot naturally share a starting position, we exchange their starting positions during generation and verify that the two conditions have identical sets of starting images. This check reduces the possibility that the starting image alone reveals the condition.

Predictions without history. Consider a complete, balanced pair with distinguishable encoded targets. A deterministic model that ignores history receives the same current observation and future action sequence for both samples and therefore produces the same prediction. This prediction cannot be strictly closer to each sample's own target than to the other target. Consequently, such a model scores at most 50% on the paired future-selection metric in Eq. 2; ties count as failures.

C. Action Delay Response Groups

The delay data retain all 11 delays from 0 to 10 simulator steps. At the five-step horizon used for the main one-step score, delays of 5–10 produce the same stationary response and are grouped together. This score therefore uses six distinguishable response groups: 0, 1, 2, 3, 4, and 5–10.

D. Dataset Statistics and Splits

D.1. Dataset Sizes

Table 3 lists the task data used in the training and evaluation comparisons. Training sizes exclude the original-environment data used for mixed training. A pair contains two trajectories with different dynamics conditions; a trajectory count refers to recorded episodes, not to the number of training windows sampled from them. An evaluation case fixes the current observation and future action sequence and can include histories and targets from several dynamics conditions.

The action delay counts refer to the dataset covering all delays from 0 to 10 simulator steps. Its Training split contains 5,120 base cases, each simulated under all 11 delays, giving 56,320 trajectories. Development and Test each contain 300 base cases, again evaluated under all 11 delays. Door Training contains 7,680 trajectories for each of the passable and blocked rules.

Several examples can share a source episode: pusher displacement can use multiple action windows from one episode, and forward and mirrored damping examples can share a source. The 10,000 gripper carry training pairs use 10,000 distinct source episodes.

D.2. Speed Values and Door Positions

Table 4 lists the speed values used for Training and evaluation. Development and Test use the same evaluation speeds but different interaction trajectories. Each split contains 300 cases for each of the 14 evaluated speeds, or 4,200 cases in total before comparing histories from different speeds. The main task results use the unseen speeds within the training range.

Door position is specified by an integer coordinate along the wall. Test uses the 42 positions $30 + 4k$, where $k = 0, \ldots, 41$. Development uses the 16 positions 65, 77, 79, 87, 112, 115, 131, 152, 167, 169, 171, 172, 175, 180, 183, 185. Training uses the integers from 36 through 188 after excluding all Development and Test positions and the additional unused positions 40, 127, and 132, leaving 96 positions. The three position sets are disjoint. Both passable and blocked rules are used across the splits; the test distinction is the door position, not a new rule.

D.3. Split Checks and Model Inputs

Split checks use source episode identifiers, trajectory template identifiers, or observation hashes, as appropriate for each task. Speed Development uses independently generated initial configurations and excludes initial states and current observations that duplicate Test cases. For Door, we also check the position sets listed above for overlap.

Hidden dynamics values, simulator states, and pair identifiers are retained for data generation, validation, and scoring. They are not passed to the prediction model, which receives only the visual observations and actions defined in Sec. 3.1.

E. Evaluation Details

Paired tasks. The strict comparisons in Eq. 2 do not count ties as correct. The two comparisons within a pair are dependent. Appendix B explains the upper bound for a deterministic model that ignores history.

Speed. For each target speed, accuracy is the proportion of cases in which matching history gives strictly lower pre-

Table 3. Dataset sizes and condition coverage.

Task	Training	Development	Test
Speed	16,384 trajectories; 32 speeds	300 cases per speed	300 cases per speed
Pusher displacement	32,768 pairs	256 pairs	256 pairs
Arm mass	32,768 pairs	256 pairs	256 pairs
Action delay	56,320 trajectories; 11 delays	300 cases $\times$ 11 delays	300 cases $\times$ 11 delays
Contact friction	32,768 pairs	256 pairs	256 pairs
Motion damping	32,768 pairs	256 pairs	256 pairs
Gripper carry rule	10,000 pairs	256 pairs	256 pairs
Door traversability	15,360 trajectories; 96 door positions	300 cases; 16 door positions	300 cases; 42 door positions
Portal exit	32,768 pairs	256 pairs	256 pairs

Table 4. Speed values in the benchmark. The four evaluation groups are used in both Development and Test. Interpolation values lie within the Training range but are absent from Training.

Setting	Speed values
Training	2.6, 2.7, 2.8, 2.9, 3.1, 3.3, 3.5, 3.7, 3.8, 3.9, 4.1, 4.2, 4.4, 4.5, 4.7, 5.0, 5.1, 5.3, 5.5, 5.7, 5.9, 6.1, 6.2, 6.3, 6.6, 6.7, 6.8, 7.0, 7.2, 7.3, 7.8, 7.9
Seen speeds	3.1, 5.1, 7.0
Interpolation	3.4, 4.8, 6.9
Lower extrapolation	1.75, 1.95, 2.15, 2.35
Upper extrapolation	8.25, 8.75, 9.5, 10.25

diction error than every alternative history. We then average accuracy equally across target speeds. Ties do not count as correct: if all histories produce the same prediction, accuracy is zero. The main results use the interpolation speeds listed in Appendix D.

Action delay. We select the candidate true future with the lowest mean squared error, resolving ties in favor of the smaller delay. The prediction is correct when the selected delay and the true delay belong to the same response group. Within each case, we first average correctness over delays in each group, then average equally across the six groups defined in Appendix C. The final accuracy is the mean over cases.

Door traversability. For a prediction made with matching history, we compare the mean squared errors to the passage and blockage futures. The prediction is correct only if the error to the true future is strictly smaller than the error to the alternative future. Accuracy is averaged over cases and the two rules; ties do not count as correct.

Multi-step prediction horizons. Each prediction step advances five simulator steps. We measure prediction errors at 5, 10, 15, 20, and 25 simulator steps after the current observation.

Mean-future reference. The reference in Eq. 3 is used only for scoring and is never provided to the model. The ratio is defined only when $\sum_q B_q > 0$. When conditions share the current observation and future actions and are equally weighted, a deterministic model that ignores history cannot achieve $E/B < 1$: it must make the same prediction across conditions, and the mean-future reference minimizes the error of such a prediction.

Response error settings. We specify the prediction horizon for each NRE analysis. The ratio in Eq. 5 is defined only when $\sum_i \|\Delta z_i\|^2 > 0$.

Uncertainty estimates. For multi-step errors, we use cluster bootstrap resampling [15, 17] at the level of source trajectories or scenes, retaining related samples together. These intervals describe scene-sampling uncertainty for the evaluated checkpoints, not variability across training runs.

Interpreting multiple metrics. The metrics describe different aspects of the same predictions and are not independent pieces of evidence.